Garrett Grossner

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Real-Time Software Engineer and Game Developer

Software Engineer specializing in game engine development and graphics programming. Experienced building custom 2D/3D engines in C++, OpenGL, GLSL, and Unity, while guiding multidisciplinary teams.

Skills

Programming Languages C++ | C | C# | OpenGL | Python | Java | HTML
Technologies Git | PyTorch | TensorFlow | Unity | Maya | ADO | Adobe CC | MS 365 | PowerSchool | SQL

Playable Titles

- ***Grappling with Ideas***, a third-person platformer focused on precise movement and time efficient level clears. **Apr 18, 2025**
- ***Lance-A-Lot***, a 2D platform fighter that limits player movement to the effective range of their weapons. **Apr 19, 2024**
- ***Stellar Sounds***, a memory puzzle game akin to Simon-says in space. **Mar 18, 2022**
- ***Space Chase***, a third-person action-adventure game featuring combat, platforming, and stealth challenges. **Mar 19, 2021**

Experience

Test Associate **Apr 2026 - Present**
Lionbridge Games | Xbox Game Studios
- Debugged and optimized gameplay systems, executed test suites and thoroughly documenting bugs with Azure DevOps.
- Validated daily changes across all modern Xbox platforms while further investigating problem areas to alleviate technical debt.

Network Analyst Assistant **Jun 2017 – Aug 2019, Oct 2025 – Dec 2025**
Kelseyville Unified School District
- Designed and launched a new digital student identification system for 800 students, with database integration through PowerSchool.
- Distributed laptops for 1500+ students, managing software distribution, network behavior, and ticketing systems with Google Admin.
- Remedied database health issues, leading frequent data recovery operations in Oracle using SQL scripts.

Game Design Teaching Assistant **Sep 2021 – Sep 2022**
California Polytechnic State University
- Collaborated with faculty to support instruction in courses covering user experience, programming, game design.
- Provided individualized and small group tutoring focused on coding challenges and game development techniques.

Projects

Software Development Engineer **Ongoing**
Project Lykos | *Custom Engine* | *Personal Project*
- Actively designing and developing a personal game engine with distinct application layering to support rapid tool development.
- Building an entity-component-system game world layer for large simulation size and memory speed up.
- Implementing a Vulkan-driven graphics pipeline for modern rendering support.

Graphics Programmer | Producer **Aug 2024 – Apr 2025**
Grappling with Ideas | *3D Puzzle-Platformer* | *Custom Engine* | *Team of 5*
- Managed a multidisciplinary team in the production cycle of a 3D game engine over 30 weeks, using C++, GLSL, and Lua scripting.
- Organized Agile sprint planning, documentation, and milestone delivery using GitHub Issues bug tracking and Gantt chart timetables.
- Integrated a keyframe animation system, including inverse kinematics, skeletal animation, path interpolation, and RK2 integration.
- Developed an instance-oriented particle system, featuring GPU driven calculations, vortex simulations, and 3D particle rendering.

Graphics Programmer | Co-Producer **Jan 2024 – Apr 2024**
Lance-A-Lot | *2D Platform Fighting Game* | *Custom Engine* | *Team of 4*
- Directed a 4-person team in building a custom 2D game engine using C++ and GLSL, all delivered within a 15-week development cycle.
- Crafted a portable BRDF graphics pipeline, with support for multi-light scenes, texture and normal mapping, and post processing.
- Implemented an integration-based physics systems, featuring support for OBB and AABB collision responses.
- Authored an event-driven UI system, consolidating scene transitions and menu states to accommodate short production timeline.

Education

Master of Science - Computer Science **Sep 2023 - Apr 2025**
DigiPen Institute of Technology

Bachelor of Science – Liberal Arts and Engineering Studies **Sep 2018 - Dec 2022**
California Polytechnic State University

Leadership and Community Involvement

Tubist - Rainbow City Performing Arts **Aug 2025 – Present**
Recording Secretary - Kappa Kappa Psi, Iota Pi Chapter **May 2021 – May 2022**
Librarian - Cal Poly Mustang Band **May 2020 – May 2022**